Specific heat

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use mass, material and temperature change to calculate heating energy.

SUCCESS CRITERIA - I can...

1. Identify ΔT rather than final temperature alone.
2. Use Q=mcΔT with matching units.
3. Compare materials through specific heat.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “specific heat” and “heat” to explain your answer. Sentence starters are optional.

specific heat: energy per mass needed for a one-degree temperature change

heat: energy transferred because of a temperature difference

temperature: a measure of thermal state; it is not an energy amount

01 Warm-up: bring back an earlier idea

A **3 kg** cart descends **5 m** from rest.
1. Use g=**10**; losses are **6 J**.
2. **Find bottom speed**.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: Q = mc ΔT; $\Delta T = T_f - T_i$.
No phase change. A temperature difference of **1°C** equals **1 K**.

Worked reasoning: A **2 kg** sample with c=**100 J**/(kg·°C) warms by **3°C**. Heat absorbed is Q=mc ΔT=**600 J**, with no phase change or work transfer in this calculation.

Your turn: Draw an energy arrow into the sample and **show which three factors produce 600 J**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two equal masses receive the same energy. Material A has a larger specific heat than B. Which has the smaller temperature rise?

Sentence starter: Both ... . They differ in ... because ... .



A: larger c



B: smaller c

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

04 Apply the physics

A heater transfers **2700 J** into a **0.20 kg** metal sample initially at **18°C**. Its specific heat is **450 J**/(kg·°C). Assume no phase change or energy loss. **Label the diagram** and **find the temperature rise** and final temperature.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why is multiplying by the final temperature instead of ΔT a mistake?

Sentence starter: The step I would change is … . I would change it because I found … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A cart starts from rest at a height of **4 m**. Can it reach **6 m** with no energy input from outside and no losses? Use kinetic energy to **explain your answer**. **Show the energy changes** and transfers. **Label units**.

Word help: kinetic energy = energy associated with motion

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

07 Exit check

A **0.50 kg** sample needs **4200 J** to warm **4°C**.
1. **Find c** and **name one thing** that must be true for this calculation to work.
2. **Label the energy arrow** and **identify what changes** in the sample.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A **0.5 kg** material absorbs **900 J** and warms **6°C**.
1. **Find c**.
2. **How much** heat warms **1.2 kg** of it by **15°C**?
3. **Label the energy arrow** and **identify what changes** in the sample.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: ① ② ③